

HUMAN-CENTERED AI COURSE PROJECT

EEG-to-Image Retrieval

Brain–vision alignment with lightweight visual adaptation

YOUR NAME YOUR STUDENT ID

YOUR TEAM MEMBERS

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Start with the human question

Can a non-invasive brain signal retrieve what a person has just seen?

Useful

Connect neural activity to interpretable visual semantics.

Constrained

Work within noisy, subject-specific EEG rather than claiming mind reading.

Responsible

Treat consent, privacy, and non-clinical use as design requirements.

Source: THINGS-EEG2; project responsible-use statement.

A difficult task with a precise metric

Subject-wise EEG decoding

- 10 participants from THINGS-EEG2
- 17 posterior EEG channels
- Repeated trials averaged within each stimulus
- 512-dimensional shared embedding space

200-way

zero-shot image retrieval

Chance performance

0.5% Top-1

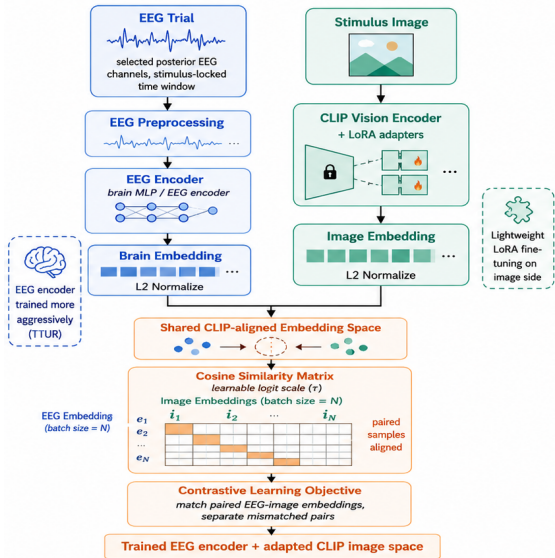
2.5% Top-5

Success: rank the exact held-out image above 199 alternatives.

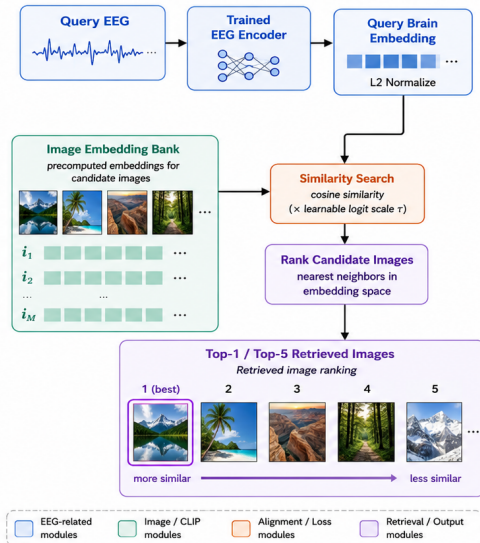
Source: Project protocol: independent cosine-similarity ranking for each EEG query.

EEG-to-Image Retrieval via Dual-Side CLIP Alignment

1. TRAINING STAGE



2. INFERENCE / RETRIEVAL STAGE



LoRA and TTUR align both modalities

Brain branch

A residual MLP maps posterior-channel EEG into the CLIP-aligned 512-D space.

Faster adaptation

$$\eta_{\text{brain}} = 5 \times 10^{-4}$$

Visual branch

Rank-32 LoRA updates CLIP ViT-B/32 without fully retraining the visual backbone.

Slower coordination

$$\eta_{\text{vision}} = 5 \times 10^{-5}$$

Contrastive alignment joins paired EEG and image embeddings.

Source: Project configuration: CLIP ViT-B/32, LoRA rank 32, TTUR-style learning rates.

Evaluation designed to resist optimism

50

independently trained
models

10 subjects $\times$ 5 seeds

- 01 Fixed final checkpoint at epoch 25
- 02 One independent ranking per query
- 03 Reloaded checkpoints reproduce metrics and predictions
- 04 No Hungarian assignment in standard results

The test set reports performance; it does not choose the checkpoint.

Source: Five-seed protocol and artifact reload checks documented in the repository.

The viewed image is retrieved reliably

86.66%

Top-1 accuracy

± 0.69 points across seeds

98.38%

Top-5 accuracy

± 0.14 points across seeds

8,666 / 10,000 Top-1 decisions and **9,838 / 10,000** Top-5 decisions were correct.

Top-1 performance is more than $170\times$ chance.

Source: Five seed-level macro averages; 200 queries per subject and seed.

Visual LoRA helps in a controlled test

All training factors are matched except frozen versus LoRA-adapted CLIP.

Controlled arm	Top-1	Top-5
Frozen CLIP SAMGA	76.17%	95.79%
SAMGA + visual LoRA/TTUR	82.68%	97.67%

+6.51 Top-1 points

95% CI: [+5.22, +7.69] | 48 of 50 paired cells improved

Source: Pre-registered paired comparison; 10,000 two-way subject/seed bootstrap resamples.

Accuracy is not enough

What the evidence supports

- Within-subject, 200-way retrieval
- Averaged repeated EEG trials
- Stable performance across five seeds

What it does not support

- Cross-subject generalization
- Single-trial mind reading
- Clinical or diagnostic interpretation

Consent, privacy, licensing, and restrained claims are part of system quality.

Source: Project limitation and responsible-use policy.

One strong result, one honest frontier

EEG can be aligned with visual semantics for reliable image retrieval.

Established

Validated retrieval: dual-side alignment, LoRA/TTUR, five seeds, and controlled attribution.

Next frontier

Retrieval-augmented reconstruction remains exploratory until its benefit is reproducible.

Performance, uncertainty, and responsible boundaries belong in the same result.

Selected references



Gifford et al. (2022).

A large and rich EEG dataset for modeling human visual object recognition.
NeuroImage.



Li et al. (2024).

Visual decoding and reconstruction via EEG embeddings with guided diffusion.
NeurIPS.



Hu et al. (2022).

LoRA: Low-rank adaptation of large language models.
ICLR.



Jiang et al. (2026).

Subject-aware multi-granularity alignment for zero-shot EEG-to-image retrieval.
arXiv preprint.